

Foundation (Jalapeno)

Q1. Simplify $\sqrt{9} + \sqrt{16}$

- (A) $3 + 4$
- (B) 5
- (C) 7
- (D) $\sqrt{25}$
- (E) $\sqrt{32}$

Q2. Simplify $\sqrt{4} \cdot \sqrt{9}$

- (A) $2 \cdot 3$
- (B) 6
- (C) 12
- (D) $\sqrt{36}$
- (E) $\sqrt{13}$

Q3. A basketball player scored $\sqrt{81}$ points in the first half. How many points did she score?

- (A) 3
- (B) 6
- (C) 9
- (D) 12
- (E) 15

Q4. Simplify $\frac{\sqrt{16}}{\sqrt{4}}$

- (A) 2
- (B) 4
- (C) $\sqrt{4}$
- (D) $\sqrt{16}$
- (E) $\frac{\sqrt{16}}{2}$

Q5. A cyclist traveled $\sqrt{25}$ miles in the first hour. How many miles did she travel?

- (A) 3
- (B) 5
- (C) 7
- (D) 9
- (E) 11

Q6. Simplify $\sqrt{9} - \sqrt{4}$

- (A) 1
- (B) 3
- (C) 5
- (D) $\sqrt{5}$
- (E) $\sqrt{13}$

Q7. A football player ran $\sqrt{36}$ yards in the first quarter. How many yards did he run?

- (A) 3
- (B) 4
- (C) 6
- (D) 8
- (E) 9

Q8. Simplify $\sqrt{16} \cdot \sqrt{9}$

- (A) 12
- (B) 24
- (C) 36
- (D) 48
- (E) $\sqrt{144}$

Open Ended Questions (FRQ)

Q9. A marathon runner's average pace is $\sqrt{16}$ miles per hour. If she runs for 2 hours, how many miles will she have run? Show your work.

Q10. Simplify the expression $\sqrt{9} + \sqrt{25} - \sqrt{16}$. Show your work.

Chef's Tips

- Encourage students to check their work.
- Remind students to simplify radicals when possible.
- Emphasize the importance of showing work for free-response questions.